

Week 7 Assignment

Virtual Power Plants (VPPs) in Advanced Distribution System Design and Operations Modeling and Scenario Analysis



WPI

In this assignment, you are required to develop a simplified analytical or simulation-based model to evaluate Virtual Power Plant (VPP) dispatch and quantify operational benefits under multiple system scenarios. You will then communicate your modeling approach, assumptions, and results in a recorded presentation.

Modeling Requirement

Develop a simplified analytical or simulation model in MATLAB, Python or Excel etc. using the parameters listed below (*The use of AI tools to assist with coding or running simulations is allowed, but you are responsible for validating results and explaining assumptions*).

Your model must evaluate VPP dispatch under three distinct scenarios (e.g., time-of-day conditions, DER availability, or system contingencies) and quantify operational impacts.

System Parameters

- Two radial underground feeders supplied from one substation
- Each feeder: 10 MW peak load and 5 MW rooftop PV
- Primary voltage: 13.2 kV (line-to-line)
- Feeder length: 5 miles (substation to end of line)
- Primary conductor: 500 kcmil underground cable
- Aggregated DER across both feeders:
 - Battery Energy Storage System (BESS): 5 MW / 20 MWh
 - Controllable EV charging load: 2 MW
- Assume balanced three-phase loading and uniform load distribution
- Voltage limits: $\pm 5\%$ of nominal voltage (0.95–1.05 pu)
- Power factor = 1.0

You may introduce additional parameters (e.g., cable resistance, efficiency, SOC limits), but they must be clearly stated and justified.

Scenarios

In your analysis simulate the following three scenarios.

1. **Evening Winter Peak (No Solar)**
Assumption: 18:00 winter peak, PV = 0 MW, EVs charging with 50% available controllable capacity.
2. **Midday High Solar Scenario**
High PV output with moderate load and partial BESS charging. No EV's available
3. **Contingency Scenario**
Choose either a Feeder fault, or DER unavailability event requiring VPP re-dispatch. Create your own assumptions.

For each scenario, perform and document the following analyses:

1. **Feeder Loading and Peak Reduction**
 - Calculate feeder and substation loading with and without VPP dispatch.
 - Quantify MW and percent peak reduction.
2. **Current and Voltage Impacts**
 - Estimate primary feeder current flows.
 - Estimate voltage drop or voltage profile improvement along the feeder (simplified calculations are acceptable).
3. **VPP Dispatch Logic**
 - Define dispatch rules for BESS and EV load control (e.g., peak shaving, load shifting, contingency support).
 - Show how state of charge (SOC) constraints affect dispatch decisions.
4. **Contingency Performance**
 - Demonstrate how the VPP mitigates system stress during the fault or constrained condition.
 - Quantify load served, curtailed, or deferred.

Create a recorded presentation (Use PPT with video/audio) (approximately 5–10 Minutes), explaining:

1. Your modeling approach and tools used.
2. All engineering and modeling assumptions.
3. Scenario definitions and dispatch logic.
4. Engineering interpretation of results (planning, operations, or reliability implications).

Clarity of explanation and engineering reasoning are critical evaluation criteria.

Extra Credit:

In your presentation discuss the distribution planning and operational impacts of Virtual Power Plants (VPPs), including implications for feeder planning, hosting capacity, non-wires alternatives, and control room operations, restoration, and system reliability. Also, address market, regulatory, and economic considerations, such as how VPPs participate in wholesale or distribution markets, major value streams (capacity, energy, and ancillary services), and regulatory or utility business model challenges. Additionally, discuss potential impacts towards cybersecurity including major cyber-physical risks associated with VPP communications and control, as well as high-level mitigation and resilience strategies.

Grading Rubric (Total Points = 200)

Category	Weight	Description
Scenario Design and VPP Dispatch Logic	35%	Well-justified scenarios and control strategies
Quantitative Results and Interpretation	35%	Quality of calculations, figures, and engineering insights
Communication and Presentation	30%	Clarity of report and video explanation